Trig, Exponentials, Logs, and Inverses
(Sections 1.3, 1.5, 1.6)

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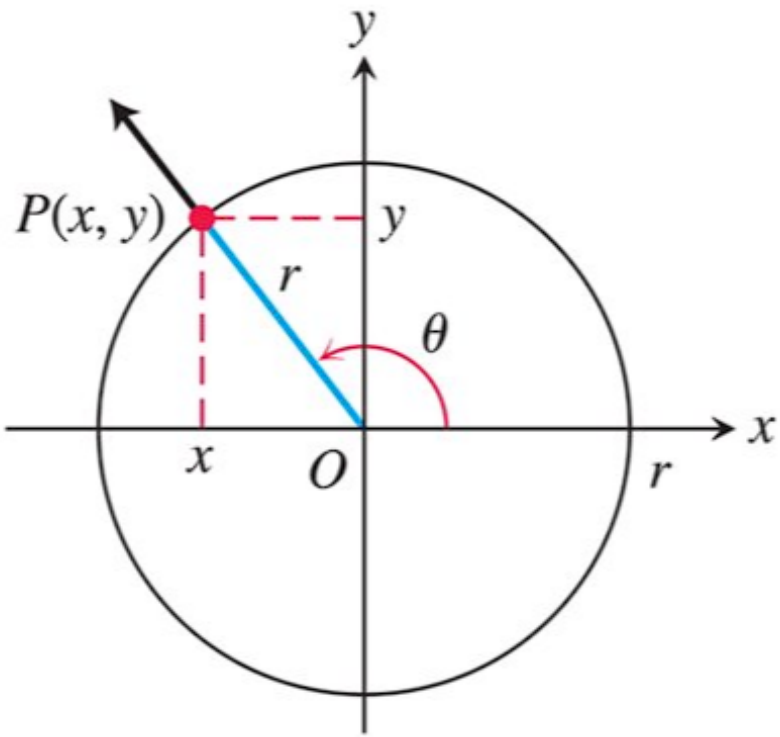
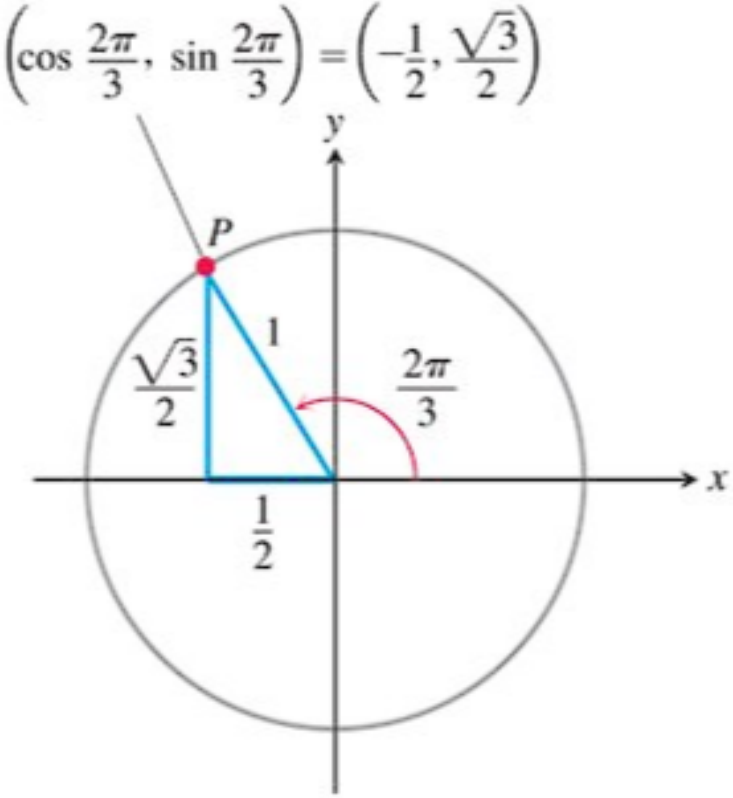
MATH 2413

The Unit Circle





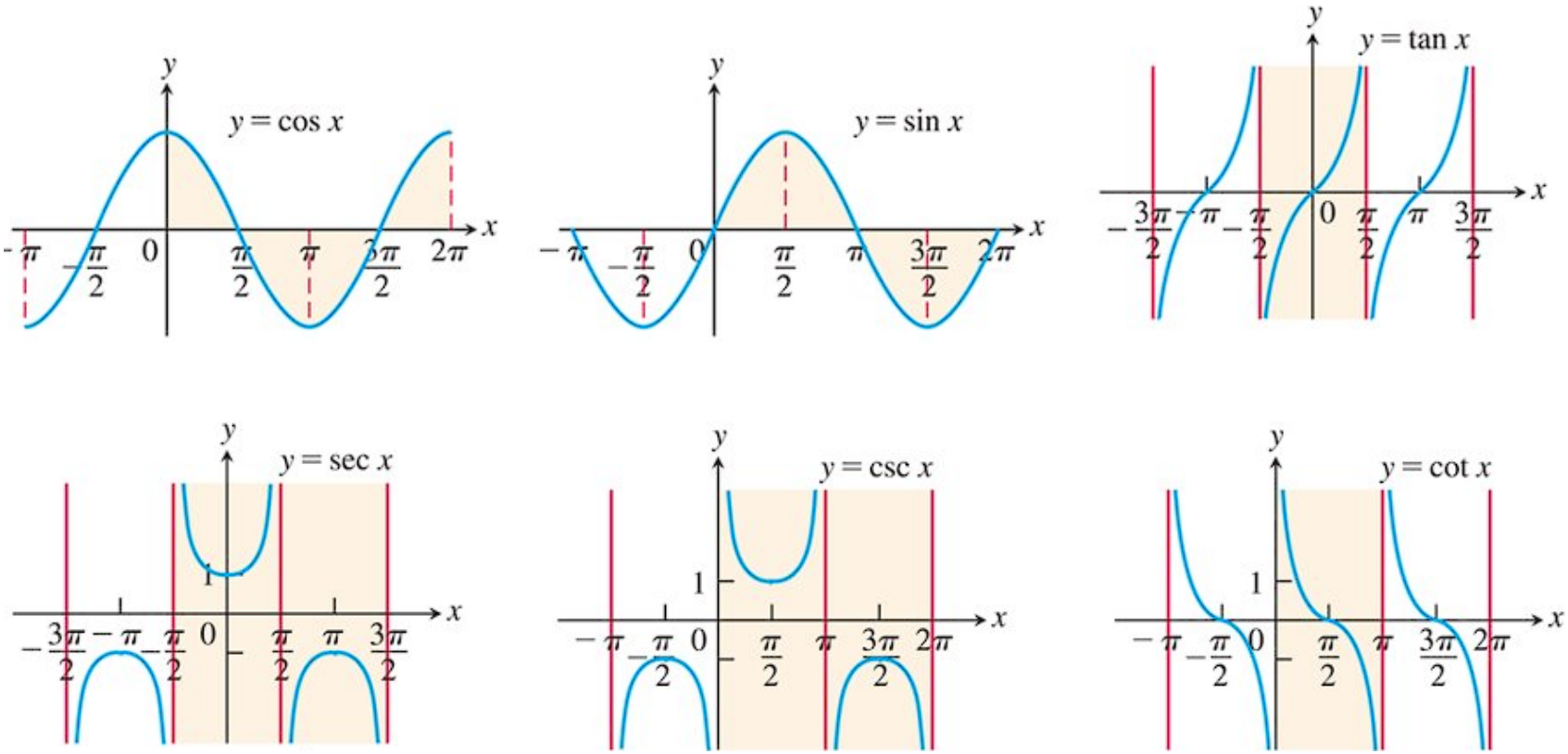
Trig Definitions & Angles

$$\sin(\theta) = \frac{y}{r} \quad \cos(\theta) = \frac{x}{r} \quad \tan(\theta) = \frac{y}{x}$$

$$\csc(\theta) = \frac{r}{y} \quad \sec(\theta) = \frac{r}{x} \quad \cot(\theta) = \frac{x}{y}$$

Degrees	-180	-135	-90	-45	0	30	45	60	90	120	135	150	180	270	360
θ (radians)	$-\pi$	$\frac{-3\pi}{4}$	$\frac{-\pi}{2}$	$\frac{-\pi}{4}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π
$\sin\theta$	0	$\frac{-\sqrt{2}}{2}$	-1	$\frac{-\sqrt{2}}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\cos\theta$	-1	$\frac{-\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{-\sqrt{2}}{2}$	$\frac{-\sqrt{3}}{2}$	-1	0	1
$\tan\theta$	0	1		-1	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$		$-\sqrt{3}$	-1	$\frac{-\sqrt{3}}{3}$	0		0

Trig Graphs



Transformations

Recall that functions can be shifted, stretched, and flipped by adding, subtracting, multiplying, and dividing constants into certain parts of the function. For example, a sine function may end up looking like this:

$$f(x) = A \sin\left(B(x - C)\right) + D$$

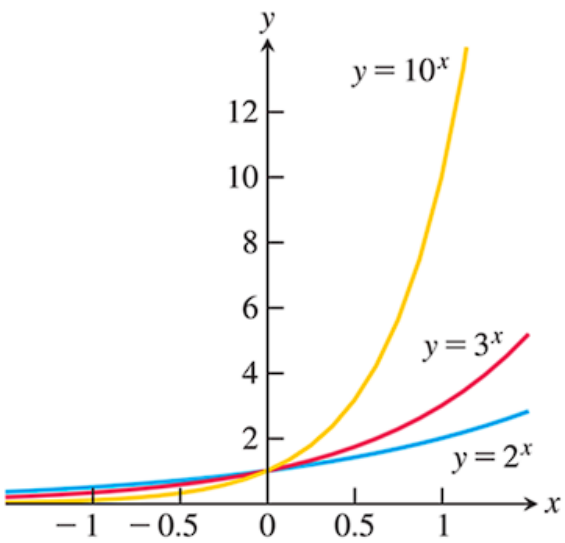
- $|A|$ is the **amplitude** (notice the absolute value). If A is negative, this represents a flip over the x axis.
- $\frac{2\pi}{|B|}$ is the **period**. If B is negative, this represents a flip over the y axis.
- C is the horizontal shift. D is the vertical shift.

Exponential Functions

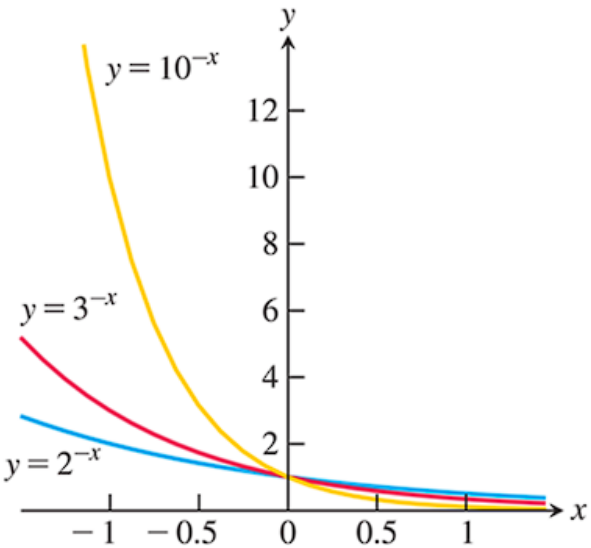
Definition

An **exponential** function is a function of the following form:

$$f(x) = a^x, \ a > 0, \ a \neq 1$$



(a) $y = 2^x$, $y = 3^x$, $y = 10^x$



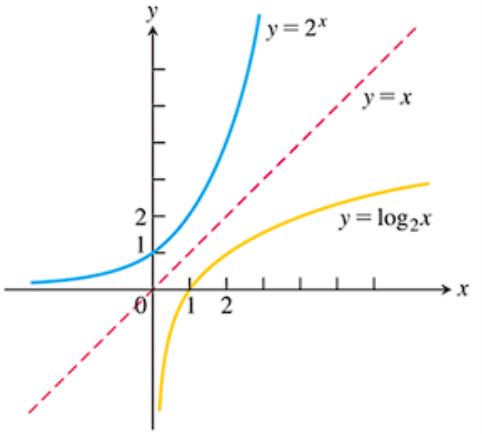
(b) $y = 2^{-x}$, $y = 3^{-x}$, $y = 10^{-x}$

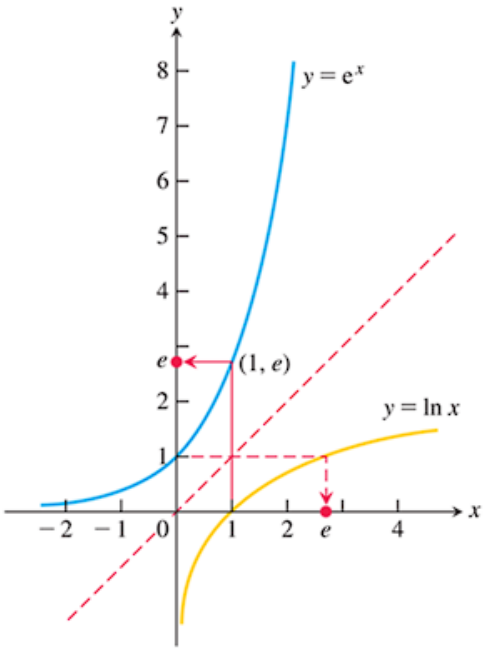
Logarithmic Functions

Definition

An **logarithmic** function, or "log", is a function of the following form:

$$f(x) = \log_a(x), \ a > 0, \ a \neq 1$$





Exponential and Log Rules

- $y = a^x \iff x = \log_a(y)$
- $y = e^x \iff x = \ln(y)$

Exponent Rule	**Log Rule**
$a^{x+y} = a^x \cdot a^y$	$\log_a(x \cdot y) = \log_a(x) + \log_a(y)$
$a^{x-y} = a^x / a^y$	$\log_a(x/y) = \log_a(x) - \log_a(y)$
$(a^x)^y = a^{x \cdot y}$	$\log_a(x^y) = y \cdot \log_a(x)$
$a^1 = a$	$\log_a(a) = 1$
$a^0 = 1$	$\log_a(1) = 0$

One-to-One Functions

Definition

An function is **one-to-one** if:

- $f(x_1) = f(x_2)$ always implies that $x_1 = x_2$
- $x_1 \neq x_2$ always implies that $f(x_1) \neq f(x_2)$
- Two different inputs cannot give the same output

(These are three equivalent definitions; they all say the same thing in different words.)

Invertible Functions

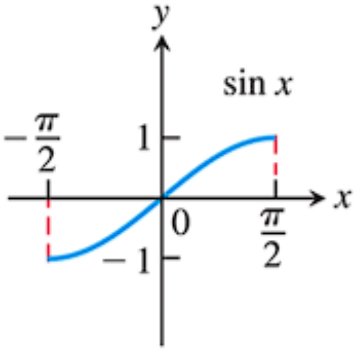
Definition

An function is **invertible** if and only if it is one-to-one.
Invertible/One-to-one functions pass the **horizontal line test**.

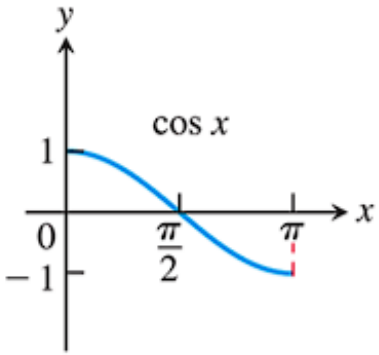
Example: $f(x) = a^x$ and $g(x) = \log_a(x)$ are inverses of each other. How do we find inverses?

Inverse Trig Functions

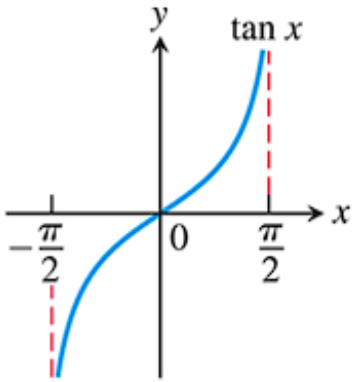
Domain restrictions that make the trigonometric functions one-to-one



$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



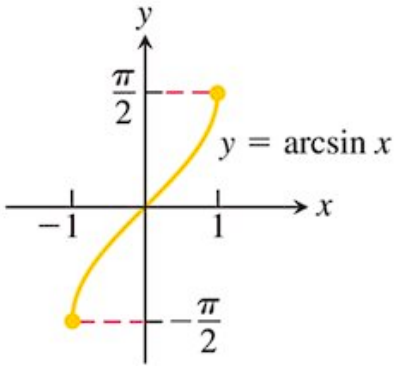
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$

Definition

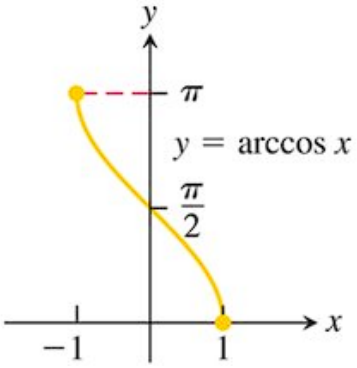
If $-1 \leq y \leq 1$, then $\arcsin(y)$ is the value of x in $[-\pi/2, \pi/2]$ such that $y = \sin(x)$.

Inverse Trig Functions

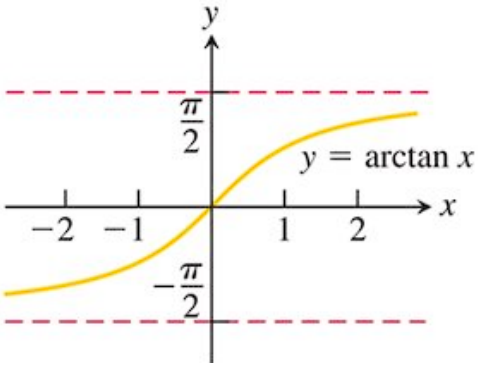
Domain: $-1 \le x \le 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



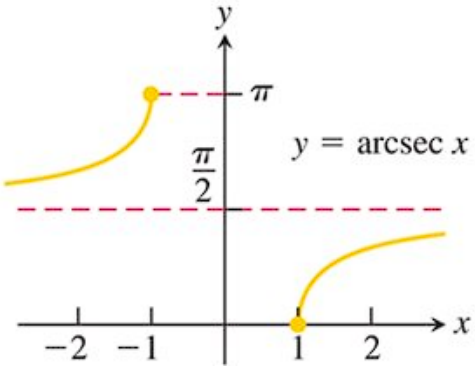
Domain: $-1 \le x \le 1$
Range: $0 \le y \le \pi$



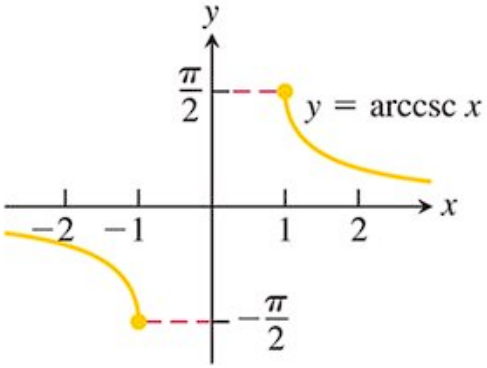
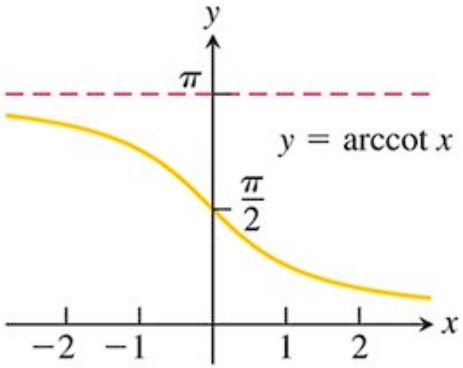
Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



Domain: $x \le -1$ or $x \ge 1$
Range: $0 \le y \le \pi, y \ne \frac{\pi}{2}$



Domain: $x \le -1$ or $x \ge 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, y \ne 0$



Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$